

Customer Support Ticket Priority Prediction and Automated Assignment System Using Agentforce

Salesforce Implementation Report

Business Domain: Customer Support Operations

Platform: Salesforce with Agentforce

Implementation Environment: Salesforce Developer Edition Org

Agentforce Agent: NM Support Agent

Subagent: Support Ticket Priority Analysis

Core Object: Support Ticket Intelligence (Support_Ticket_Intelligence__c)

Automation: Auto-Launched Flow – Support Ticket Intelligence

Project Repository: github.com/mohanraj3010/Nm-Agentforce-support-Flow

Overview

The Customer Support Ticket Priority Prediction and Automated Assignment System is a Salesforce-based solution that reduces manual effort in support-ticket handling. It analyzes the description of a customer's latest support ticket, determines whether the ticket is High, Medium or Low priority, returns the support level to which the ticket is assigned, and creates an urgent handling task when the priority is High. Support users interact with the solution conversationally through Agentforce, while the business automation is executed by a Salesforce Flow.

The solution is built entirely with declarative Salesforce configuration. Its components are one custom object (**Support Ticket Intelligence**), one **Auto-Launched Flow**, and an Agentforce configuration made up of the **NM Support Agent**, the **Support Ticket Priority Analysis** subagent and the **Analyze Support Ticket Priority** action. The standard Account, Contact, User and Task objects support the solution. No Apex code, Lightning web components or external services are part of the implementation.

How Priority Is Determined

In this report, “priority prediction” means the automatic determination of a ticket's priority level from the text of its description. The determination follows keyword conditions configured in the Flow. **No independent machine-learning model was trained.** Agentforce provides the conversational interface and invokes the Flow; the Flow performs the classification, task creation and assignment.

Table 1: Implementation Status

Component	Status
Support Ticket Intelligence object and fields	Completed
Auto-Launched Flow: Support Ticket Intelligence	Completed
Agentforce agent, subagent and action configuration	Completed
Runtime testing in the Developer Edition org (five scenarios)	Completed – all scenarios passed

Documentation Conventions

Implementation Note callouts state the exact scope of the implementation. Figures are screenshots captured from the org. Names appear exactly as configured.

Objectives

The primary objective of the project is to provide a Salesforce-based solution that analyzes customer support tickets, classifies their priority and initiates the appropriate support handling automatically. The objectives that follow from it are:

- Automatically classify support tickets as High, Medium or Low priority.
- Retrieve the latest support ticket associated with a customer Account.
- Reduce the manual effort involved in prioritizing and assigning tickets.
- Create an urgent handling task for High-priority tickets.
- Use Agentforce to analyze ticket details and trigger the backend automation.
- Support faster ticket response and resolution handling.

PHASE 1: IDEATION & REQUIREMENT ANALYSIS

This phase identifies the problem faced by support teams, records how the affected users experience it, compares the solution options considered, and defines the requirements that the implementation had to satisfy.

1.1 Problem Identification

Customer support teams receive a large number of tickets every day, while prioritization and assignment are often performed manually. Manual handling is time-consuming, can be inconsistent, and can leave critical tickets waiting behind lower-priority ones. The challenges identified were a high volume of tickets, manual prioritization, assignment to the wrong agent or team, delayed resolution, and a poor customer experience when urgent issues are not handled quickly.

Table 2: Problem Statements

PS	I am	I'm trying to	But	Because	Which makes me feel
PS-1	Support Agent	focus on critical tickets first	tickets are not always prioritized automatically	ticket prioritization can be manual	frustrated
PS-2	Customer	get urgent issues resolved quickly	critical issues may wait behind lower-priority tickets	support teams handle large ticket volumes	concerned
PS-3	Support Manager	assign tickets to the right support level	assignment may require manual decisions	ticket urgency is not consistently analyzed	overwhelmed
PS-4	Support Agent	identify urgent issues from descriptions	important keywords can be missed	ticket descriptions contain different urgency signals	uncertain
PS-5	Support Manager	reduce SLA-related risk	older unresolved tickets may remain unnoticed	SLA risk needs additional checking	concerned

1.2 Empathy Map

The empathy map was prepared for the primary user persona, the Support Agent.

Table 3: Empathy Map – Support Agent

Dimension	Observation
Think & Feel	Wants critical issues identified quickly so that urgent customer problems receive attention.
Hear	Receives customer issue descriptions and support requests.
See	Ticket status, issue type, description, priority level, assignment and SLA risk.
Say & Do	Reviews tickets, handles assigned cases and focuses on urgent support issues.
Pain	Manual prioritization, missed urgency signals and delays in handling critical tickets.
Gain	Automatic priority classification, clearer assignment, urgent-task creation and reduced manual effort.

1.3 Brainstorming and Solution Selection

Four options were considered for managing support tickets. Salesforce CRM with Agentforce was selected.

Table 4: Solution Options Considered

Option	Outcome
Manual ticket registers	Not selected
Spreadsheet-based ticket tracking	Not selected
Standalone support portal	Not selected
Salesforce CRM with Agentforce	Selected

Salesforce CRM was selected because it provides:

- Automation through Salesforce Flow.
- Ticket analysis through Agentforce.
- Automatic priority classification.
- Automatic urgent-task creation.
- Reduced manual support effort.

The brainstorming session also produced further ideas, such as real-time notifications for urgent tickets, SLA breach alerts, and dashboards for ticket trends. These were recognized as valuable extensions outside the scope of the completed implementation and are recorded as future enhancements.

1.4 Functional Scope

The project builds a ticket-analysis capability on Salesforce in which a support user supplies an Account Name, the latest ticket for that Account is retrieved and analyzed, and the outcome is returned conversationally.

Major Functional Areas

- Custom object: Support Ticket Intelligence, which stores ticket details.
- Association of tickets with Accounts and Contacts.
- Auto-Launched Flow for retrieval, description analysis, priority classification, assignment and message generation.
- Urgent Task creation for High-priority tickets.
- Agentforce subagent and action for conversational interaction.

Table 5: Salesforce Components Implemented

Category	Component
Custom object	Support Ticket Intelligence (Support_Ticket_Intelligence__c)
Standard objects used	Account, Contact, User, Task
Automation	Auto-Launched Flow “Support Ticket Intelligence” using Get Records, Assignment, Decision and Create Records elements

Category	Component
Agentforce	NM Support Agent; subagent Support Ticket Priority Analysis (Support_Ticket_Priority_Analysis); Flow-based action Analyze Support Ticket Priority
Programmatic components	None. No Apex classes, triggers, Lightning web components or Visualforce pages are used.
Version control	Public GitHub repository: Nm-Agentforce-support-Flow

1.5 Business Objectives

The delivery objectives that map directly to implemented components are:

- Store ticket details in a single custom object with a unique Ticket ID (Support Ticket Intelligence).
- Identify the latest ticket for a customer Account (Get Account and Get Ticket elements of the Flow).
- Classify ticket priority from the description (Analyze Description decision).
- Create an urgent handling task for High-priority tickets (Create Records element, Task object).
- Return the support level to which the ticket is assigned (varAssignedTo).
- Provide a conversational interface for the analysis (Support Ticket Priority Analysis subagent and its action).

1.6 Stakeholder Mapping

Table 6: Stakeholders and Responsibilities

Stakeholder	Role in the Solution
Customer	Raises support issues and expects urgent problems to be resolved quickly.
Support Agent	Reviews and handles tickets, and focuses on urgent issues. Uses the analysis outcome to see priority and assignment.
Support Manager	Needs tickets assigned to the right support level and consistent urgency analysis.
System Administrator	Configures the Salesforce environment, object, Flow and Agentforce components.

1.7 Customer Journey

The customer journey follows a ticket from creation to resolution. The first stage records the ticket in Salesforce; the next six stages are performed by Agentforce and the Flow; the final two stages are carried out by the support team.

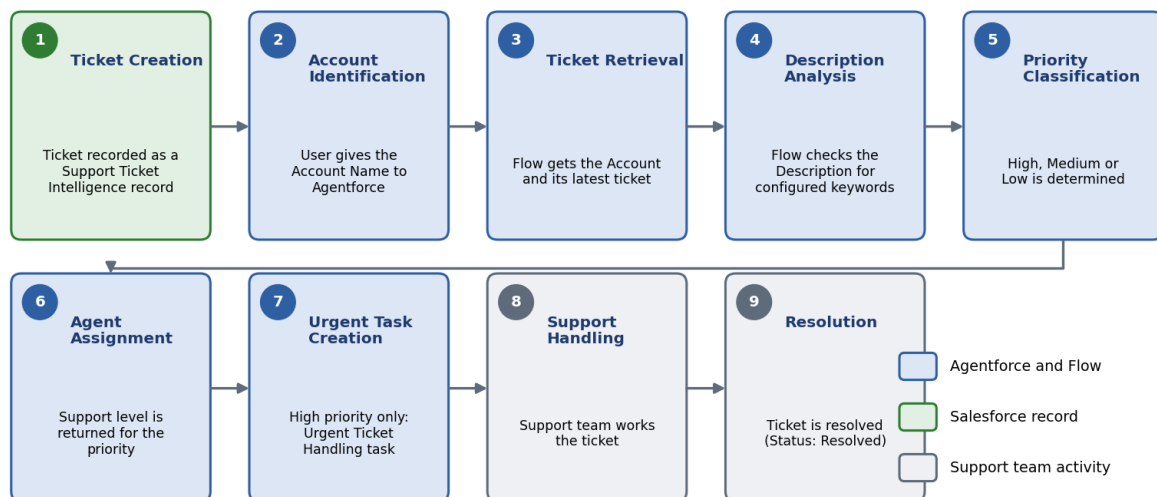


Figure 1: Customer journey from ticket creation to resolution

Table 7: Customer Journey Stages

Stage	Description	Salesforce / Agentforce Element
1. Ticket Creation	A customer issue is recorded as a ticket.	Support Ticket Intelligence record (Ticket ID TKT-{0000})
2. Account Identification	The support user provides the Account Name.	Agentforce input varAccountName; Get Account element
3. Ticket Retrieval	The latest ticket for the Account is retrieved.	Get Ticket element (CreatedDate descending, first record)
4. Description Analysis	The ticket description is checked against configured conditions.	Analyze Description decision
5. Priority Classification	The ticket is classified as High, Medium or Low.	varPriorityLevel
6. Agent Assignment	The support level for the priority is returned.	varAssignedTo
7. Urgent Task Creation	For High priority, an urgent handling task is created.	Create Records element (Task)
8. Support Handling	The support team handles the ticket.	Ticket Status field (New, In Progress)
9. Resolution	The ticket is resolved.	Ticket Status field (Resolved); Resolution Time field

1.8 Solution Requirements

Table 8: Functional Requirements

FR No	Requirement	Realized By
FR-1	Support ticket creation and management	Support Ticket Intelligence object with Ticket ID, Issue Type, Description, Priority Level, Status, Created Date, Assigned To, SLA Breach Risk and Resolution Time
FR-2	Account and Contact association	Customer (Lookup to Account) and Contact (Lookup to Contact) fields

FR No	Requirement	Realized By
FR-3	Ticket description analysis	Analyze Description decision in the Flow
FR-4	Automatic priority classification	Keyword conditions that result in High, Medium or Low (default)
FR-5	High-priority task creation	Create Records element creating the “Urgent Ticket Handling” Task
FR-6	Support agent assignment	varAssignedTo set to Senior Support Agent, Support Agent Tier 2 or Support Agent Tier 1
FR-7	SLA breach risk checking (optional capability)	SLA Breach Risk checkbox available on the object; see Section 4.6 for scope
FR-8	Agentforce conversational analysis	Support Ticket Priority Analysis subagent with the Analyze Support Ticket Priority action

Table 9: Non-Functional Requirements

NFR	Description	How It Is Addressed
Usability	Clear Agentforce interaction and understandable priority messages	Agentforce asks for the Account Name when it is missing; configured messages describe the outcome in plain language
Reliability	Consistent priority classification based on configured conditions	Fixed keyword conditions in the Flow decision
Performance	Quick retrieval and analysis of the latest support ticket	Get Records elements return only the first record, sorted by CreatedDate descending
Security	Controlled Salesforce access to customer and ticket information	See Section 6.4
Maintainability	Flow and Agentforce configuration can be updated	Declarative configuration with no custom code
Scalability	Supports increasing support-ticket volumes through Salesforce automation	Processing is performed by Salesforce Flow on the platform

1.9 Data Flow

Two flows of data are involved. The ticket data flow moves from the support request to the final response. The Agentforce flow moves from the user’s Account Name to the response shown in the conversation.

Table 10: Data Flow Stages

No.	Data Flow	Description
1	Support request → Account	The Account Name provided by the user identifies the customer Account.
2	Account → latest ticket	The latest Support Ticket Intelligence record for the Account is selected.
3	Ticket → description	The ticket Description text is read.
4	Description → priority level	The description is analyzed to produce High, Medium or Low.
5	Priority → assignment and task	The support level is assigned; a Task is created for High priority.

No.	Data Flow	Description
6	Result → Agentforce → user	The five output variables are returned to Agentforce, which responds to the user.

1.10 Technology Stack

Table 11: Technology Stack

Layer	Technology
Platform	Salesforce
Database	Support_Ticket_Intelligence__c with Account, Contact, User and Task records
Logic	Auto-Launched Flow
Conversational layer	Agentforce
Agent configuration	Support Ticket Priority Analysis subagent
Automation elements	Flow Decision, Assignment and Create Records
Output	Agentforce conversation response
Version control	Git / GitHub repository

1.11 Data Governance and Integrity

Table 12: Data Governance and Integrity Mechanisms

Mechanism	Implementation
Unique ticket identifier	The standard Name field is an Auto Number field with the format TKT-{0000}.
Controlled values	Issue Type, Priority Level and Status are picklists with fixed values.
Relationship integrity	Customer, Contact and Assigned To are Lookup fields to Account, Contact and User.
Latest-ticket selection	The Flow sorts tickets by CreatedDate descending and uses the first record only.
Agent scope control	The subagent is limited to ticket analysis, priority determination and triggering backend automation; billing, subscription management and account updates are excluded.
Controlled actions	The subagent is instructed not to update records directly or send emails and to use the Flow for all actions.

PHASE 2: PROJECT DESIGN & PLANNING

This phase maps the identified problems to solution features, defines the architecture and data model, and records how the work was planned.

2.1 Problem–Solution Fit

Table 13: Problem–Solution Fit

Problem	Solution	Benefit
Manual ticket prioritization	The Flow analyzes the description and sets the priority	Faster identification of urgent cases
Urgent issues may be missed	High-priority keyword decision	Critical cases receive attention
Manual task creation	The Flow creates a Task for High priority	Reduced manual effort
Manual support assignment	The Flow sets the assigned support level	More consistent handling
SLA risk can be overlooked	SLA Breach Risk field on the ticket; optional SLA risk decision	Better risk awareness (see Section 4.6 for scope)
Need for conversational access	Agentforce subagent	Interaction through conversation

2.2 Solution Summary

Table 14: Solution Summary

Parameter	Description
Problem	Manual ticket prioritization and assignment cause delays in handling critical customer support issues.
Solution	A Salesforce-based Customer Support Ticket Priority Prediction and Automated Assignment System.
Approach	Agentforce combined with an Auto-Launched Flow to analyze ticket descriptions, classify priority, assign a support level and trigger urgent-task creation.
Impact	Faster identification of critical tickets, reduced manual effort, and better support-team productivity.
Business context	Customer support operations involving customers, support agents, support managers and Agentforce-assisted ticket handling.
Scalability	Supports increasing ticket volumes through Salesforce automation, with scope for additional assignment rules, SLA monitoring and priority indicators.

2.3 Solution Architecture

The architecture has two layers. In the conversational layer, a support user talks to the NM Support Agent. The agent routes the request to the Support Ticket Priority Analysis subagent, which invokes the Analyze Support Ticket Priority action. In the Salesforce automation and data layer, that action runs the Support Ticket Intelligence Flow, which reads Account and ticket records, applies the priority logic and, for High priority, creates a Task. The Flow’s output variables travel back through Agentforce to the user.

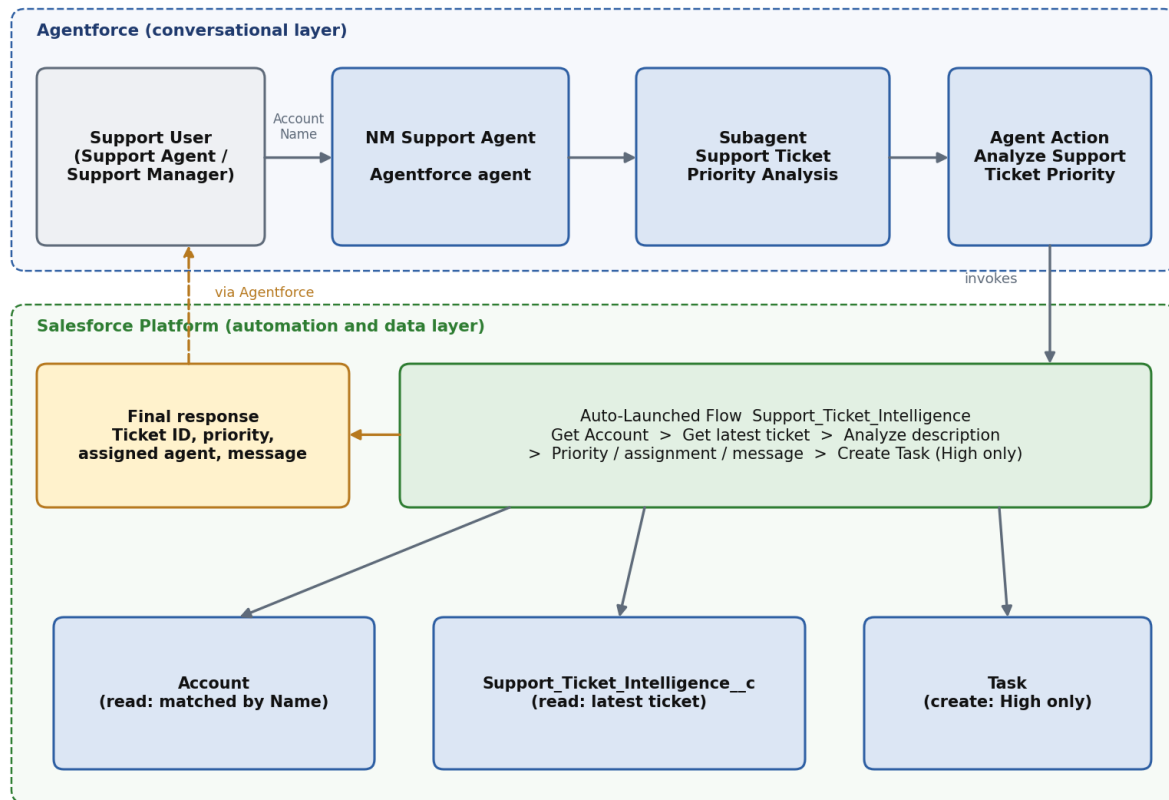


Figure 2: Solution architecture

Table 15: Architecture Layers

Layer	Component	Responsibility
Conversational	NM Support Agent	Interacts with the support user
Conversational	Support Ticket Priority Analysis subagent	Handles ticket-priority requests within its defined scope and follows the configured instructions
Conversational	Analyze Support Ticket Priority action	Passes the Account Name to the Flow and receives the output variables
Automation	Support Ticket Intelligence Flow	Retrieves data, classifies priority, assigns the support level, creates the Task and sets the message
Data	Account, Support_Ticket_Intelligence__c, Task	Hold the customer, ticket and urgent-task records

2.4 Salesforce Data Model

The Support Ticket Intelligence object is the centre of the data model. Each ticket belongs to an Account through the Customer lookup, may identify a Contact, and may reference the User assigned to it. For High-priority tickets the Flow creates a Task related to the ticket.

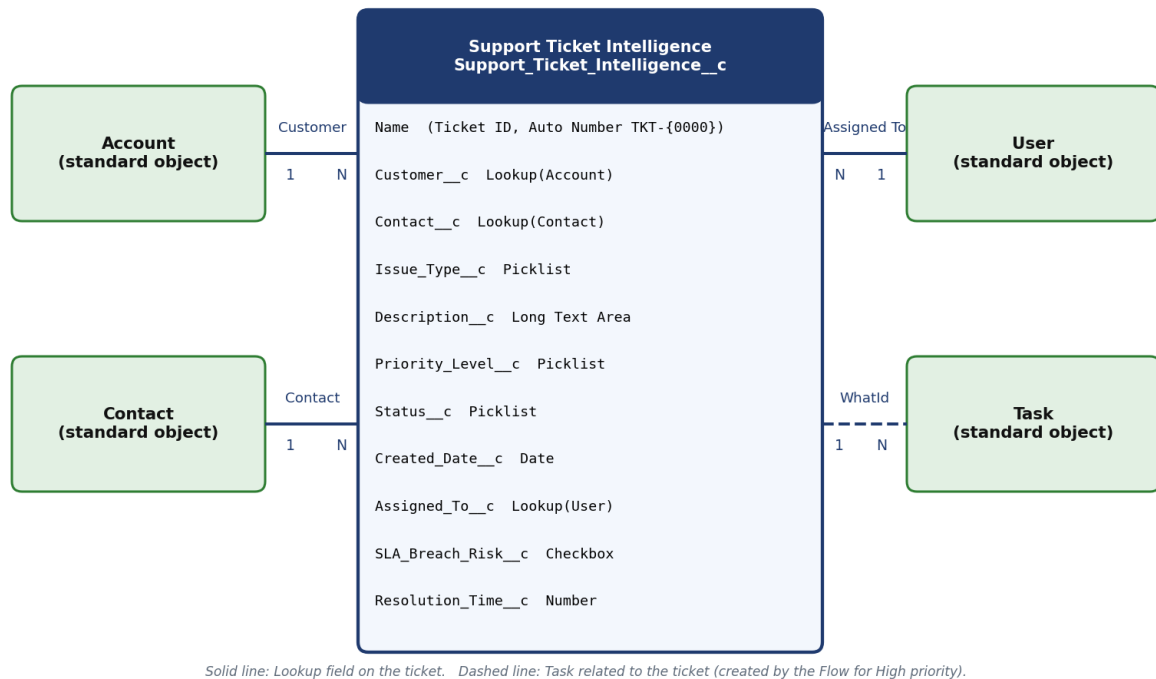


Figure 3: Salesforce data model

Table 16: Relationships

Relationship	Type	Field Label
Support Ticket Intelligence → Account	Lookup	Customer
Support Ticket Intelligence → Contact	Lookup	Contact
Support Ticket Intelligence → User	Lookup	Assigned To
Task → Support Ticket Intelligence	Related To (WhatId), set by the Flow	Related To

2.5 Project Planning

The project followed an Agile approach: epics were broken into user stories, stories were estimated in story points, and work was executed in sprints.

Execution Roadmap

1. **Developer setup** – configure the Salesforce environment.
2. **Data modeling** – create the Support Ticket Intelligence object, fields and relationships; associate tickets with Accounts and Contacts.
3. **Automation** – build the Auto-Launched Flow to retrieve the latest ticket, analyze its description and classify priority.
4. **Agentforce and assignment** – configure the subagent and action; create the urgent task and assign the support level for High priority.
5. **SLA management** – SLA-related field and optional risk check.
6. **Conversational support** – return Ticket ID, priority, assigned agent and action message through Agentforce.

Table 17: Product Backlog

Sprint	Epic	USN	User Story	Pts	Priority
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to configure the Salesforce environment for Support Ticket Intelligence so that the support automation solution can be developed.	3	High
Sprint-2	Data Modeling	USN-2	As an admin, I want to create the Support Ticket Intelligence object with required fields and relationships so that customer support ticket information is stored systematically.	5	High
Sprint-2	Data Modeling	USN-3	As a support user, I want support tickets to be associated with Accounts and Contacts so that the latest customer ticket can be identified for analysis.	3	High
Sprint-3	Automation	USN-4	As a support agent, I want an Auto-Launched Flow to retrieve the latest ticket and analyze its description so that ticket priority can be determined automatically.	5	High
Sprint-3	Automation	USN-5	As a support agent, I want tickets to be automatically classified as High, Medium or Low based on configured urgency conditions so that critical issues receive attention first.	5	High
Sprint-4	Agentforce & Automation	USN-6	As an AI Agent (Agentforce), I want to analyze support ticket details and trigger the backend Flow so that manual ticket prioritization is reduced.	8	High
Sprint-4	Task & Assignment	USN-7	As a support manager, I want High-priority tickets to automatically create an urgent handling task and assign the appropriate support level so that critical tickets are handled quickly.	5	High
Sprint-5	SLA Management	USN-8	As a support manager, I want the system to check SLA breach risk for older unresolved tickets so that potential SLA issues can be identified.	5	Medium
Sprint-6	Agentforce & Conversational Support	USN-9	As a support user, I want to provide an Account Name to Agentforce and receive the ticket ID, priority, assigned agent and action message so that I can understand the ticket handling outcome conversationally.	5	Medium

The backlog totals 44 story points across six sprints. USN-8 is addressed through the SLA Breach Risk field and the optional SLA check described in Section 4.6; comprehensive SLA monitoring is not part of the implementation.

PHASE 3: SALESFORCE ENVIRONMENT & DATA MODEL IMPLEMENTATION

This phase documents the Salesforce environment and the implementation of the custom object that stores support tickets.

3.1 Salesforce Environment Setup

A Salesforce Developer Edition org was created through the Salesforce Developer sign-up page and used for the entire implementation.

1. Open the Salesforce Developer sign-up page and complete the form: first and last name, email, Role (Developer), Company (college name), Country (India), Postal Code, and a username in the format username@organization.com.
2. Activate the account using the verification email that Salesforce sends to the registered address.
3. Set a password and answer a security question; Salesforce then redirects the user to the Setup home page.

3.2 Custom Object: Support Ticket Intelligence

Table 18: Custom Object Definition

Attribute	Detail
Object Label	Support Ticket Intelligence
API Name	Support_Ticket_Intelligence__c
Record Name (Ticket ID)	Standard Name field, type Auto Number, format TKT-{0000}
Purpose	Stores support ticket details: customer, contact, issue type, description, priority, status, assignment, SLA breach risk and resolution time.

Implementation Note: The visible Ticket ID is the object's standard **Name** field, configured as an Auto Number with the format TKT-{0000} (for example TKT-0002). The implementation does not use a separate Ticket_Number__c field.

3.3 Object Fields

Table 19: Support Ticket Intelligence Fields

No.	Field Label	API Name	Data Type	Values / Notes
1	Ticket ID	Name	Auto Number	Format TKT-{0000}
2	Customer	Customer__c	Lookup (Account)	Related Account
3	Contact	Contact__c	Lookup (Contact)	Customer contact
4	Issue Type	Issue_Type__c	Picklist	Technical, Billing, General
5	Description	Description__c	Long Text Area	Issue details; analyzed by the Flow
6	Priority Level	Priority_Level__c	Picklist	Low, Medium, High
7	Status	Status__c	Picklist	New, In Progress, Resolved
8	Created Date	Created_Date__c	Date	Ticket date

No.	Field Label	API Name	Data Type	Values / Notes
9	Assigned To	Assigned_To__c	Lookup (User)	Support agent
10	SLA Breach Risk	SLA_Breach_Risk__c	Checkbox	Risk flag
11	Resolution Time	Resolution_Time__c	Number	Time taken to resolve the ticket (hours)

Implementation Note: The custom **Created Date** field (Created_Date__c, type Date) records the ticket date. The Flow selects the latest ticket using the Salesforce system field **CreatedDate**, not this custom field.

Fields & Relationships			
14 Items, Sorted by Field Label			
FIELD LABEL	FIELD NAME	DATA TYPE	
Assigned To	Assigned_To__c	Lookup(User)	
Contact	Contact__c	Lookup(Contact)	
Created By	CreatedById	Lookup(User)	
Created Date	Created_Date__c	Date	
Customer	Customer__c	Lookup(Account)	
Description	Description__c	Long Text Area(32768)	
Issue Type	Issue_Type__c	Picklist	
Last Modified By	LastModifiedById	Lookup(User)	

Figure 4: Support Ticket Intelligence fields in Object Manager

3.4 Standard Objects Used

Table 20: Standard Objects and Their Role

Object	Role in the Solution
Account	Represents the customer. The Flow finds the Account whose Name equals the value supplied by the user; tickets point to it through the Customer lookup.
Contact	Identifies the customer contact for a ticket through the Contact lookup.
User	Referenced by the Assigned To lookup field on the ticket.
Task	Created by the Flow for High-priority tickets and related to the ticket.

PHASE 4: AUTOMATION & BUSINESS LOGIC

The business logic of the solution is implemented in a single Auto-Launched Flow. The Flow has no record trigger; it runs only when it is invoked from Agentforce, and it returns its results to the caller through output variables.

4.1 Flow Overview

Table 21: Flow Overview

Attribute	Configuration
Flow Label	Support Ticket Intelligence
Flow API Name	Support_Ticket_Intelligence
Flow Type	Auto-Launched Flow (no trigger; invoked from Agentforce)
Input	varAccountName
Outputs	varAccountId, varTicketId, varPriorityLevel, varAssignedTo, varActionMessage
Invoked By	Agentforce action Analyze Support Ticket Priority
Elements Used	Get Records, Assignment, Decision, Create Records

4.2 Flow Variables

Table 22: Flow Variables

Label	API Name	Type	Direction	Purpose
Account Name	varAccountName	Text	Input	Name of the customer Account supplied by the user
Account Id	varAccountId	Text	Output	Id of the Account that was found
Ticket Id	varTicketId	Text	Output	Id of the latest ticket that was analyzed
Priority Level	varPriorityLevel	Text	Output	Calculated priority: High, Medium or Low
Assigned Agent	varAssignedTo	Text	Output	Support level assigned to the ticket
Action Message	varActionMessage	Text	Output	Final message describing the priority and the action taken

4.3 Flow Logic

The Flow executes the following sequence for every invocation.

Table 23: Flow Steps

Step	Flow Activity	Detail
1	Receive Account Name	The input variable varAccountName receives the Account Name from Agentforce.
2	Get Account	Get Records on Account where Name equals varAccountName; first record only. The Account Id is stored in varAccountId.
3	Get latest ticket	Get Records on Support_Ticket_Intelligence__c where Customer__c equals varAccountId, sorted by CreatedDate descending.

Step	Flow Activity	Detail
4	Select latest ticket	Only the first record is stored, so the most recent ticket is used. Its Id is stored in varTicketId.
5	Check ticket exists	The Flow checks whether a ticket was returned. If none exists, no priority analysis is performed and Agentforce reports that no support ticket was found.
6	Analyze Description	A Decision element evaluates the ticket Description using CONTAINS conditions (see Section 4.4).
7	Determine priority	The matching outcome sets varPriorityLevel to High, Medium or Low (default).
8	Priority / assignment handling	High: create the urgent Task and assign Senior Support Agent. Medium: assign Support Agent Tier 2. Low: assign Support Agent Tier 1. The matching message is stored in varActionMessage.
9	Return result	The output variables are returned to Agentforce.

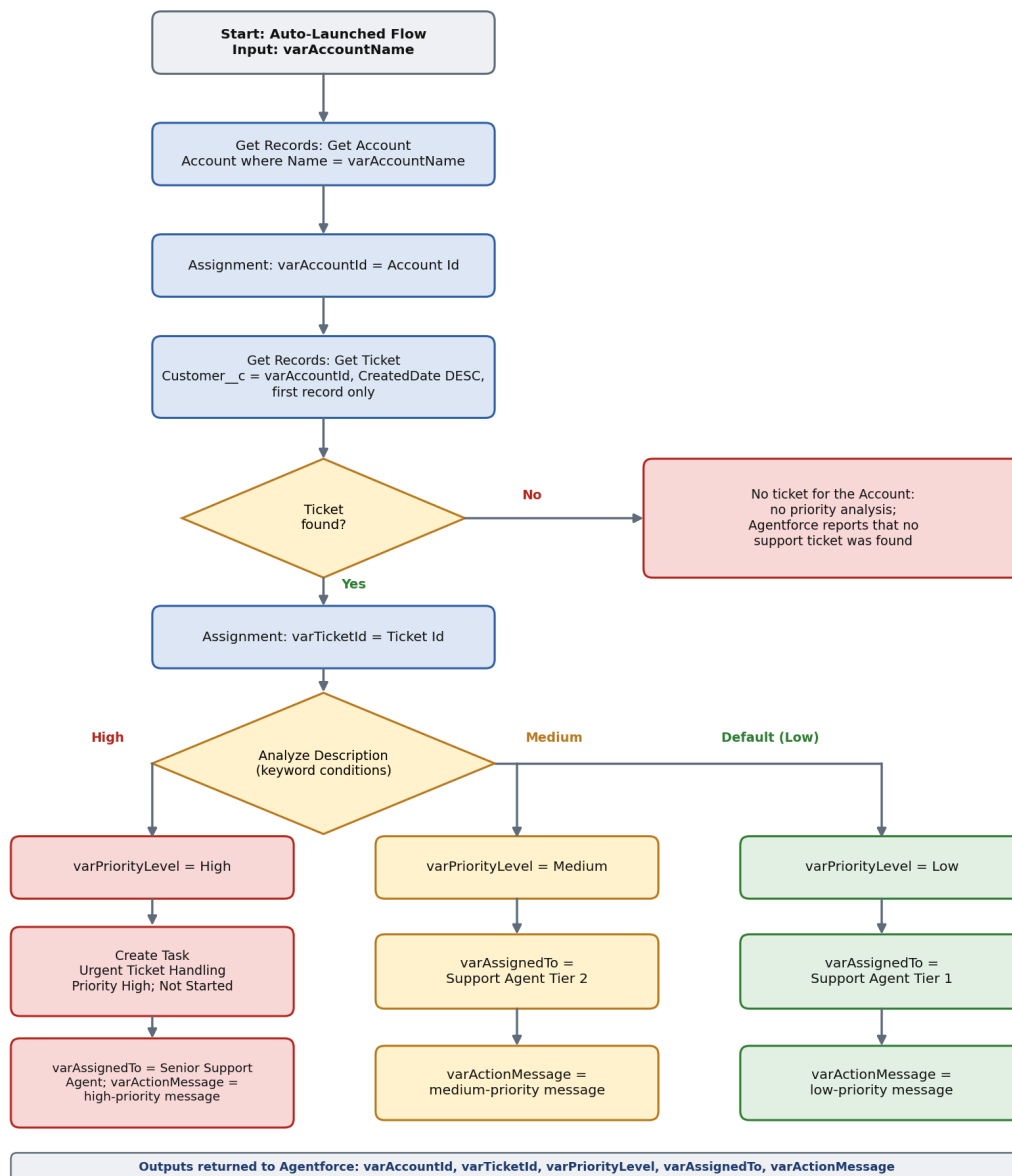


Figure 5: Logic of the Support Ticket Intelligence Flow

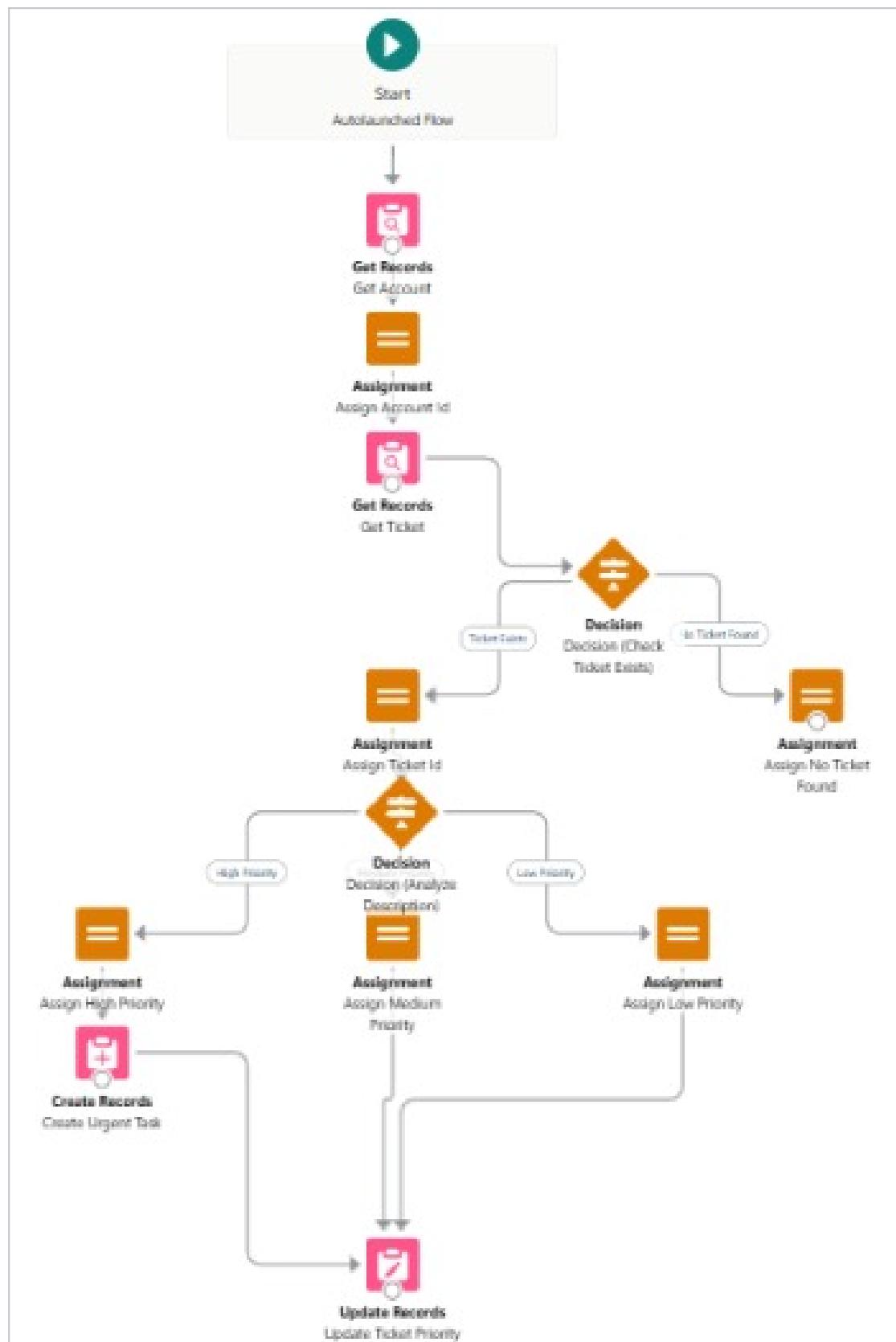


Figure 6: Support Ticket Intelligence Flow in Flow Builder

4.4 Priority Classification Rules

The Analyze Description decision uses CONTAINS conditions on the ticket Description. High-priority keywords are evaluated first, then Medium-priority keywords. A description that matches neither set takes the default outcome, Low.

Table 24: Priority Classification Rules

Priority	Description Contains	Assigned To	Additional Action	Configured Message
High	“urgent”, “not working” or “failure”	Senior Support Agent	Create Task “Urgent Ticket Handling”	High priority ticket detected. Assigned to senior agent.
Medium	“issue”, “slow” or “delay”	Support Agent Tier 2	None	Ticket marked as medium priority. Will be handled shortly.
Low	None of the above (default outcome)	Support Agent Tier 1	None	Tickets are low priority and queued for processing.

High-priority keywords are evaluated first, then Medium; any other description is Low.

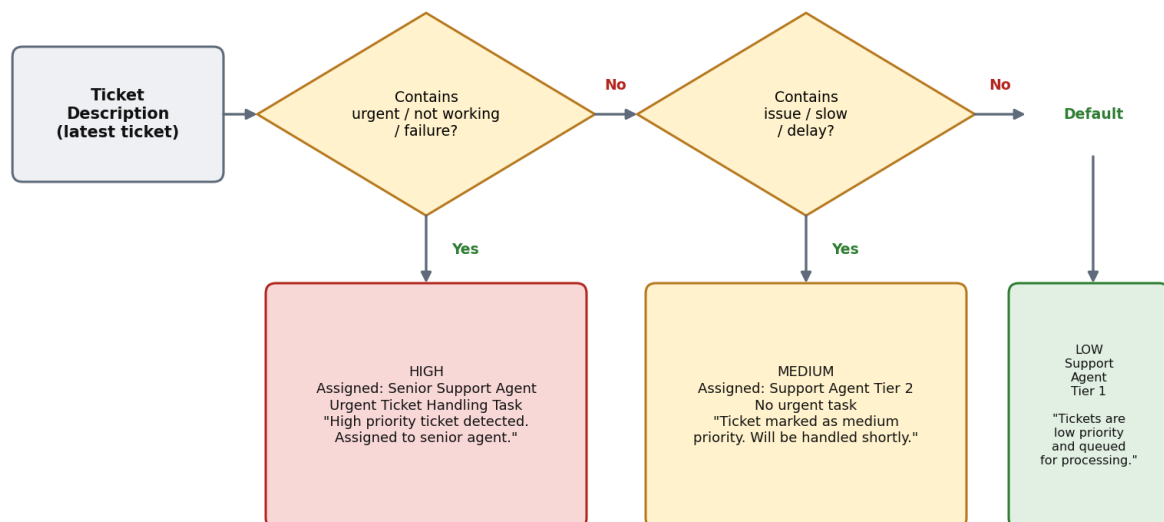


Figure 7: Priority classification logic

4.5 High-Priority Ticket Processing

When the description matches a High-priority keyword, the Flow sets the priority, creates an urgent handling Task related to the ticket, assigns the Senior Support Agent level and sets the high-priority message.

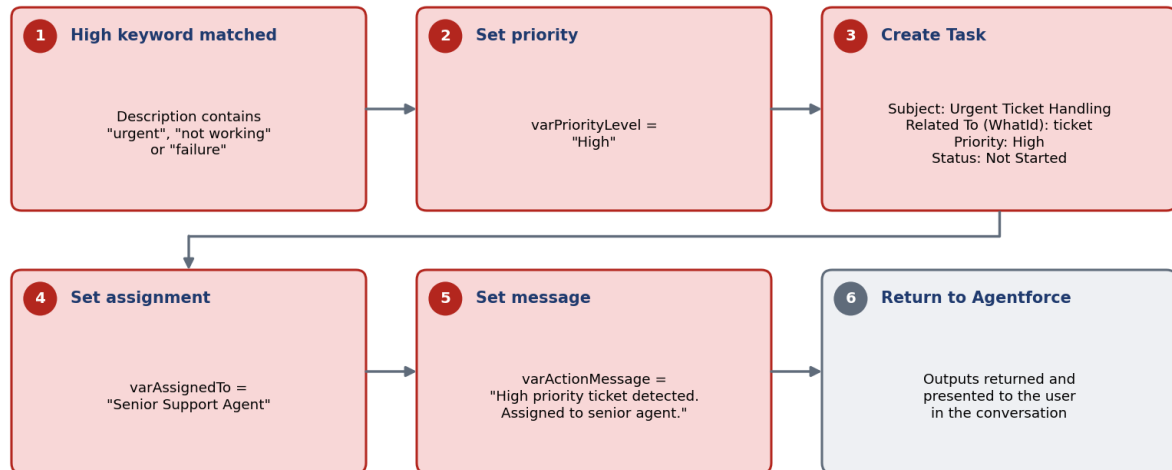


Figure 8: High-priority ticket processing

Table 25: Urgent Task Configuration

Task Field	Value
Subject	Urgent Ticket Handling
Related To (WhatId)	{!varTicketId}, the ticket that was analyzed
Priority	High
Status	Not Started

Implementation Note: The Flow returns the assigned support level as text in **varAssignedTo** (Senior Support Agent, Support Agent Tier 2 or Support Agent Tier 1). This report does not state that the Flow writes the value into the ticket's Assigned To lookup field.

4.6 SLA Breach Risk – Scope

Table 26: SLA Breach Risk Scope

Aspect	Status
SLA Breach Risk field	Available. SLA_Breach_Risk__c is a checkbox on the Support Ticket Intelligence object.
SLA risk check	Described in the project design as optional, with the example condition “Created Date older than 2 days”. It is not part of the priority and assignment logic validated in the runtime tests.
Automated SLA monitoring and escalation	Not implemented.
Classification in this report	Optional capability; current limitation; future enhancement.

PHASE 5: AGENTFORCE CONFIGURATION

Agentforce provides the conversational entry point to the solution. This phase documents the agent, the subagent that handles ticket-priority requests, and the action that connects the subagent to the Flow.

5.1 Role of Agentforce and the Flow

Table 27: Division of Responsibility

Component	Role	Responsibilities
Agentforce	Conversational interaction and orchestration	Asks for the Account Name when it is missing; passes it to the Flow through the action; presents the returned Ticket ID, priority, assigned agent and message to the user; reports when no ticket is found.
Salesforce Flow	Configured business automation	Retrieves the Account and latest ticket; analyzes the description; determines priority; creates the Task for High priority; sets the assigned support level and the message; returns the outputs.

Implementation Note: No machine-learning model was trained for this project. Priority is determined by the keyword conditions configured in the Flow. Agentforce does not decide the priority on its own; it invokes the Flow and presents the result.

5.2 Agent Setup

1. Enable Agentforce in the org (Salesforce Go → Agentforce → Get Started → Turn On → Confirm).
2. Open Agentforce Agents in Setup and open the agent in Agentforce Builder.
3. In the Subagents panel, choose New → New Subagent and enter the name, API name, classification description, scope and instructions.
4. Open the subagent's actions, choose Create New Action, select Reference Action Type "Flow" and select the Support Ticket Intelligence Flow.
5. Configure the action description, loading text, input and outputs, and finish the action.

Agentforce Builder requires the agent to be deactivated before subagents or actions are added or changed.

5.3 Agent and Subagent Configuration

Table 28: Agent and Subagent Configuration

Attribute	Configuration
Agent	NM Support Agent
Subagent Name	Support Ticket Priority Analysis
Subagent API Name	Support_Ticket_Priority_Analysis
Classification Description	Analyzes customer support tickets to determine priority based on issue description and urgency. Handles user requests related to ticket status, issue severity and support prioritization.

Attribute	Configuration
Scope	Only analyzes support ticket descriptions, determines the priority level (High, Medium, Low) and triggers backend automation for task assignment. Does not handle unrelated requests such as billing, subscription management or account updates.
Action	Analyze Support Ticket Priority (Flow: Support_Ticket_Intelligence)

5.4 Subagent Instructions

The subagent is configured with the following instructions, which it follows whenever the topic is triggered:

1. Ask the user to provide the Account Name if it has not already been given.
2. Retrieve the latest support ticket associated with the Account.
3. Read and analyze the ticket description.
4. Determine priority from keywords: High if the description contains “urgent”, “not working” or “failure”; Medium if it contains “issue”, “slow” or “delay”; otherwise Low.
5. Set the priority level accordingly.
6. If the ticket is High priority, trigger the Auto-Launched Flow and ensure a task is created for immediate handling.
7. Assign the ticket to the appropriate support level (for example, a senior agent for High priority).
8. Return a clear response message: High – urgent handling is initiated; Medium – the issue will be handled shortly; Low – the ticket is queued.
9. Do not update records directly or send emails; use the Flow for all actions.



Support Ticket Priority Analysis

Analyzes customer support tickets to determine priority based on issue description and urgency.
Handles ticket status, ticket severity and ticket prioritization.

Reasoning Instructions ❶

1. Ask the user to provide the Account Name if it has not been provided.
2. Call the action Analyze Support Ticket Priority using the provided Account Name.
3. Do not predict or assign ticket priority directly; strictly rely on the Flow's output.
4. Present the Ticket ID, Priority Level, Assigned Agent, and the Status Message clearly to the user.
5. If no ticket or account is found, inform the user clearly and ask them to verify the Account Name.
6. Do not perform any other CRM action, send emails, or draft responses outside ticket priority analysis.

Actions Available For Reasoning ❶

Figure 9: Support Ticket Priority Analysis subagent in Agentforce Builder

5.5 Agent Action: Analyze Support Ticket Priority

Table 29: Agent Action Configuration

Attribute	Configuration
Action Label	Analyze Support Ticket Priority
Reference Action Type	Flow
Reference Action	Support_Ticket_Intelligence
Action Description	Analyzes customer support tickets based on the provided account name, determines ticket priority using issue description, and triggers automation to assign tasks for high-priority cases.
Loading Text	Analyzing ticket details and determining priority...

Table 30: Action Input and Outputs

Name	Direction	Description	Data Type	Settings
varAccountName	Input	Enter the customer account name to fetch the latest support ticket and analyze its priority.	lightning__textType	Require input: enabled Collect data from user: enabled
varAccountId	Output	Returns the unique identifier of the customer account used to retrieve the support ticket.	lightning__textType	Show in conversation: enabled Rendering: Text
varTicketId	Output	Returns the unique identifier of the support ticket analyzed by the system.	lightning__textType	Show in conversation: enabled Rendering: Text
varPriorityLevel	Output	Represents the calculated ticket priority level (High, Medium, Low) based on the issue description.	lightning__textType	Show in conversation: enabled Rendering: Text
varAssignedTo	Output	Indicates the support agent or team assigned to handle the ticket.	lightning__textType	Show in conversation: enabled Rendering: Text
varActionMessage	Output	Displays the final message indicating ticket priority and actions taken by the system.	lightning__textType	Show in conversation: enabled Rendering: Text

varAccountName String
* Description ⓘ
The exact name of the customer Account.

Advanced Settings

✓ Require Input to execute action ⓘ

Outputs are the results the agent provides after completing an action

varAccountId String

* Description ⓘ
The Salesforce ID of the Account record.

Advanced Settings

☐ Filter from agent context ⓘ

☐ Show in conversation ⓘ

varActionMessage

String

* Description ⓘ
Status and action message explaining the priority decision

Advanced Settings

☐

Filter from agent context ⓘ

☒ Show in conversation ⓘ

varAssignedTo String

* Description ⓘ
The designated support agent or tier assigned to handle this ticket.

Advanced Settings

☐ Filter from agent context ⓘ

☒ Show in conversation ⓘ

varPriorityLevel String

* Description ⓘ

Calculated priority level for the support ticket (High, Medium, or Low).

Advanced Settings

Figure 10: Analyze Support Ticket Priority action configuration

5.6 Conversational Behaviour

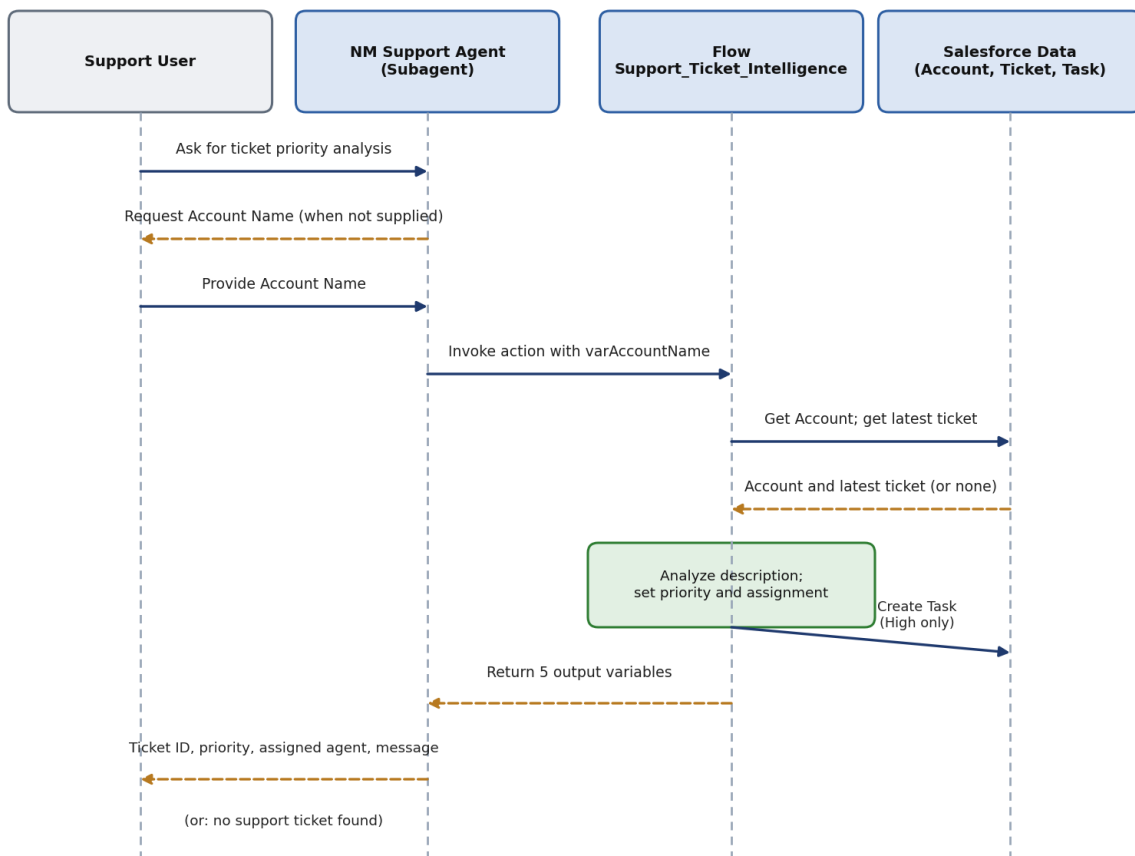


Figure 11: Agentforce interaction sequence

Table 31: Agentforce Behaviour by Scenario

Scenario	Agentforce Behaviour
Account Name not provided	Asks the user to provide the Account Name.
Account Name provided; ticket found	Invokes the Flow and returns the Ticket ID, priority, assigned agent and action message.
High priority	Returns the high-priority message; the Flow creates the urgent Task and assigns Senior Support Agent.
Medium priority	Returns the medium-priority message; assigned to Support Agent Tier 2.
Low priority	Returns the low-priority message; assigned to Support Agent Tier 1.
Account has no support ticket	Reports that no support ticket was found.
Unrelated request (billing, subscription management, account updates)	Outside the scope of the subagent and not handled.

PHASE 6: TESTING, VALIDATION & SECURITY

6.1 Testing Approach

Runtime testing was performed by conversing with the agent in Agentforce against ticket data held in the org. Five scenarios were executed: one for each priority level, and two exception scenarios (missing Account Name and an Account without a support ticket). Test Accounts NM Acme Corporation, NM Beta Industries and NM Gamma Traders each had a support ticket; NM Epsilon Labs had none.

6.2 Runtime Test Cases and Results

Table 32: Runtime Test Cases

Test Case	Type	Input	Expected Result	Actual Result	Status
TC-01	High priority	Account: NM Acme Corporation	High priority; Senior Support Agent; high-priority message	High priority; Senior Support Agent; high-priority status message returned	PASS
TC-02	Medium priority	Account: NM Beta Industries	Medium priority; Support Agent Tier 2; medium-priority message	Medium priority; Support Agent Tier 2; medium-priority status message returned	PASS
TC-03	Low priority	Account: NM Gamma Traders	Low priority; Support Agent Tier 1; low-priority message	Low priority; Support Agent Tier 1; low-priority status message returned	PASS
TC-04	Missing Account Name	“Check the latest ticket”	Agentforce asks for the Account Name	Agentforce asked the user to provide the Account Name	PASS
TC-05	No support ticket	Account: NM Epsilon Labs	Agentforce reports that no ticket was found	Agentforce reported that no support ticket was found	PASS

Table 33: Test Summary

Total	Passed	Failed
5	5	0

Table 34: Requirement Coverage

Requirement	Covered By	Remarks
FR-1, FR-2	TC-01 to TC-03, TC-05	Tickets and Accounts used as test data; no-ticket case verified
FR-3, FR-4	TC-01 to TC-03	High, Medium and Low classification verified
FR-5	TC-01	High-priority path exercised; Task creation is configured in the Flow (Section 4.5)
FR-6	TC-01 to TC-03	Support level returned for each priority
FR-7	Not tested	Optional capability; see Section 4.6

Requirement	Covered By	Remarks
FR-8	TC-01 to TC-05	Conversational behaviour verified, including both exception scenarios

6.3 Runtime Evidence

The following frames hold the runtime screenshots captured for each test case and the Agentforce execution trace.

Live Test [Dropdown] [Info]

Hi, I'm an AI service assistant. How can I help you?

Check the latest ticket for NM Acme Corporation

The latest support ticket for NM Acme Corporation is high priority. It has been assigned to a Senior Support Agent.

- Ticket ID: a00g700002GjLVuAAN
- Priority Level: High
- Assigned Agent: Senior Support Agent
- Status: High priority ticket detected. Assigned to senior agent.

Would you like more details about this ticket or need help with another account?

Agent Preview Details

Context Variables [5]

Summary

Input: Input

Transition to Subagent: Support Ticket Priority Analysis

Reasoning: Support Ticket Priority Analysis

Action: Analyze Support Ticket Priority

Reasoning: Support Ticket Priority Analysis

Output Evaluation: GROUNDED

Figure 12: TC-01: High-priority result for NM Acme Corporation

Live Test [Dropdown] [Info]

Would you like more details about this ticket or need help with another account?

Check the latest ticket for NM Beta Industries

Here are the details for the latest support ticket for NM Beta Industries:

- Ticket ID: a00g700002GjLVvAAN
- Priority Level: Medium
- Assigned Agent: Support Agent Tier 2
- Status: Ticket marked as medium priority. Will be handled shortly.

Would you like to check another account or need more details about this ticket?

Agent Preview Details

Context Variables [5]

Summary

Input: Input

Transition to Subagent: Support Ticket Priority Analysis

Reasoning: Support Ticket Priority Analysis

Action: Analyze Support Ticket Priority

Reasoning: Support Ticket Priority Analysis

Output Evaluation: GROUNDED

Figure 13: TC-02: Medium-priority result for NM Beta Industries

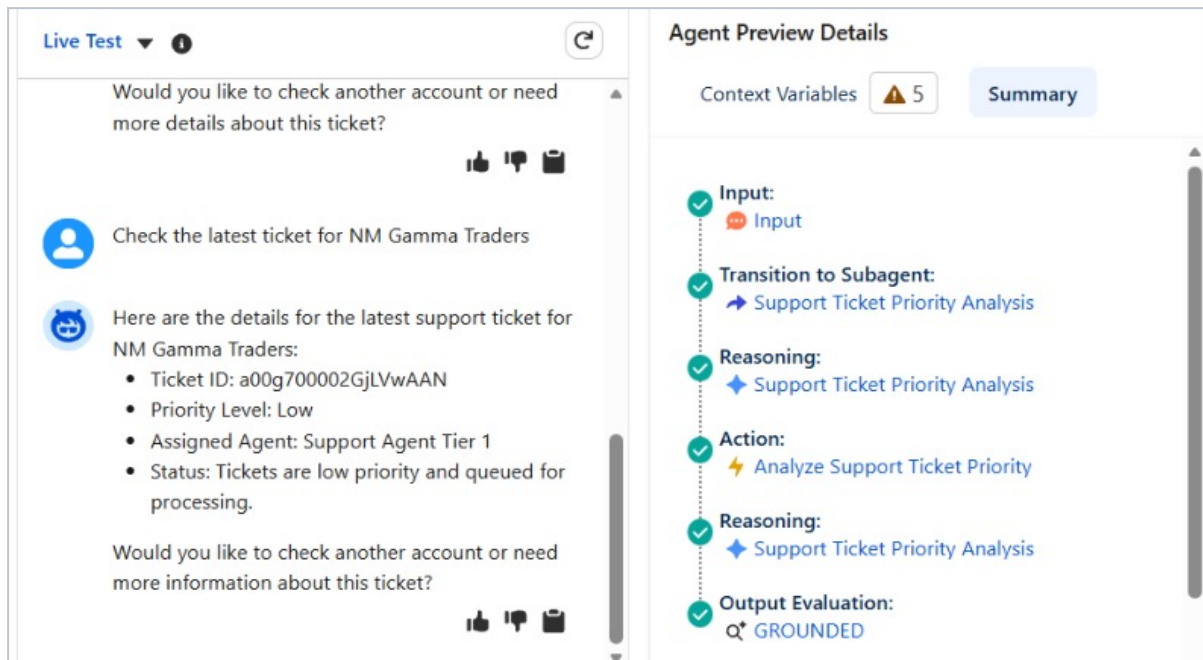


Figure 14: TC-03: Low-priority result for NM Gamma Traders

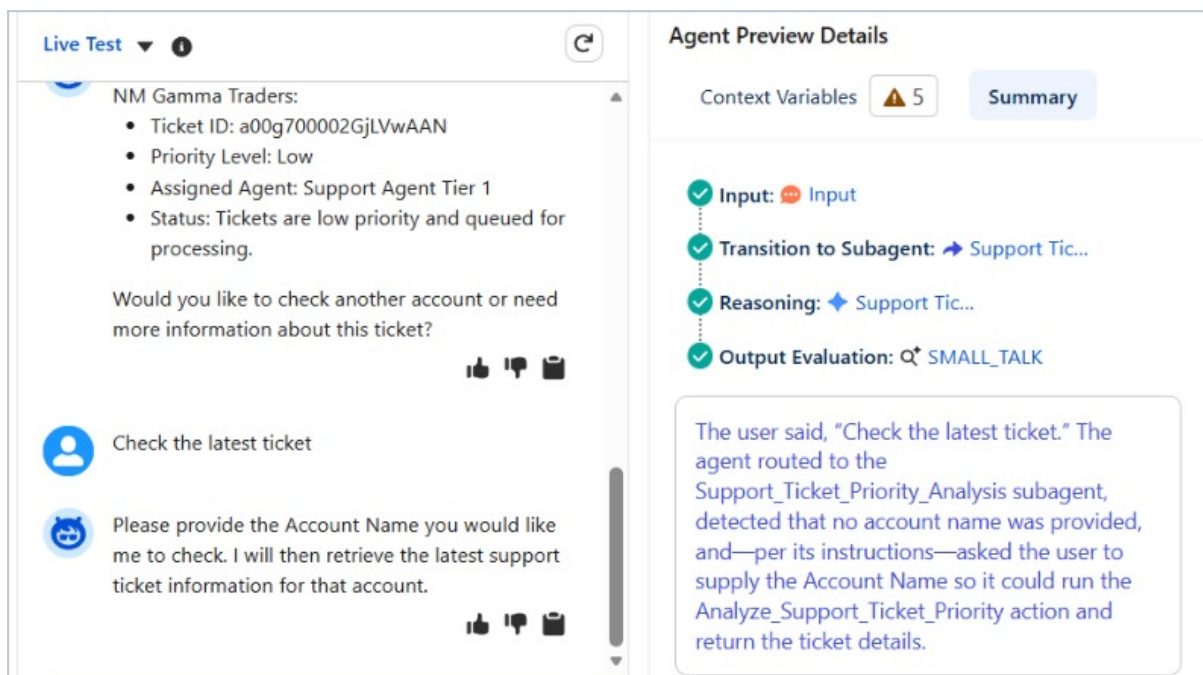


Figure 15: TC-04: Agentforce requests the Account Name

Live Test

Check the latest ticket

Please provide the Account Name you would like me to check. I will then retrieve the latest support ticket information for that account.

Check the latest ticket for NM Epsilon Labs

I could not find a support ticket for NM Epsilon Labs. Please check the Account Name and try again.
Would you like to search for another account or need further assistance?

Agent Preview Details

Context Variables 5

Summary

Input:

Input

Transition to Subagent:

Support Ticket Priority Analysis

Reasoning:

Support Ticket Priority Analysis

Action:

Analyze Support Ticket Priority

Reasoning:

Support Ticket Priority Analysis

Output Evaluation:

GROUNDED

Figure 16: TC-05: No support ticket found for NM Epsilon Labs

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Agent Preview Details

Context Variables

⚠ 5

Summary

- ✓ **Input:**
💬 Input
- ✓ **Transition to Subagent:**
➡ Support Ticket Priority Analysis
- ✓ **Reasoning:**
💠 Support Ticket Priority Analysis
- ✓ **Action:**
⚡ Analyze Support Ticket Priority
- ✓ **Reasoning:**
💠 Support Ticket Priority Analysis
- ✓ **Output Evaluation:**
🔍 GROUNDED

Figure 17: Agentforce Trace Showing Support Ticket Priority Analysis and Flow Action Execution

6.4 Security and Access Control

Security information is limited to what the project configuration supports. The tables distinguish between access requirements stated in the project documentation and configuration that is verified.

Table 35: Documented Access Requirement

Requirement	Source
Controlled Salesforce access to customer and ticket information	Non-functional requirement (Security), Section 1.8

Table 36: Verified Configuration

Control	Configuration
Agent scope	The subagent handles only ticket-description analysis, priority determination and triggering backend automation. Billing, subscription management and account updates are excluded.
Controlled actions	The subagent is instructed not to update records directly or send emails and to use the Flow for all actions.
Required input	The Account Name is a required input that Agentforce collects from the user.
Outputs shown in conversation	Only the five Flow outputs configured on the action are shown: Account Id, Ticket Id, priority level, assigned agent and action message.

Implementation Note: The project documentation does not name specific profiles, permission sets or sharing settings for this solution. None are stated or assumed in this report; access to the object and Flow follows the org's standard Salesforce security configuration.

PHASE 7: DEPLOYMENT, VERSION CONTROL & MAINTENANCE

7.1 Implementation Environment

The solution was built, configured and tested in a Salesforce Developer Edition org. No separate production release or automated deployment pipeline is part of this project. The components below make up the solution.

Table 37: Solution Component Inventory

Component	Type	Name
Custom object	Salesforce object	Support Ticket Intelligence (Support_Ticket_Intelligence__c)
Flow	Auto-Launched Flow	Support Ticket Intelligence (Support_Ticket_Intelligence)
Agent	Agentforce agent	NM Support Agent
Subagent	Agentforce subagent	Support Ticket Priority Analysis (Support_Ticket_Priority_Analysis)
Action	Agentforce action (Flow)	Analyze Support Ticket Priority

7.2 Version Control

Table 38: Project Repository

Attribute	Detail
Repository	Nm-Agentforce-support-Flow
URL	https://github.com/mohanraj3010/Nm-Agentforce-support-Flow.git
Visibility	Public
Purpose	Project repository and version control for the Salesforce implementation
CI/CD	Not used

7.3 Maintenance

Table 39: Maintenance Activities

Area	Maintenance Activity
Priority keywords	Update the Analyze Description conditions in the Flow if the keywords that signal urgency change.
Assignment logic	Update the support-level values set in the Flow if support teams change, and add rules if more teams are needed.
Flow outputs	When variables are added to the Flow, configure them on the Agentforce action so that they reach the conversation.
Agent instructions	Deactivate the agent before editing subagents or actions, then reactivate it. Keep the instructions consistent with the Flow logic.

Area	Maintenance Activity
Ticket data quality	Keep the Customer lookup and Description populated so that tickets can be found and analyzed.

7.4 Monitoring and Troubleshooting

Table 40: Troubleshooting Guide

Symptom	Likely Cause	What to Check
Agentforce keeps asking for the Account Name	No Account Name provided in the request	Provide the Account Name; check that the input is configured as required and collected from the user
No support ticket found	No ticket has the Account in its Customer field, or the Account Name does not match	Check the Account Name spelling and the Customer lookup on the ticket
Unexpected priority	The latest ticket's description differs from the expected keywords	Review the description of the latest ticket and the keyword conditions
An output is missing in the conversation	Output not configured for display	Check "Show in conversation" on the action output
A configuration change is not possible	The agent is active	Deactivate the agent, make the change and reactivate

The guide is derived from the configured components and logic; it is guidance for maintenance rather than a record of executed procedures.

7.5 Documentation Summary

Table 41: Documentation Summary

Area	Summary
Data Architecture	One custom object, Support_Ticket_Intelligence__c, with eleven fields and lookups to Account, Contact and User.
Business Process	Ticket retrieval for an Account, description analysis, priority classification, support-level assignment and urgent-task creation for High priority.
Automation	One Auto-Launched Flow with Get Records, Assignment, Decision and Create Records elements.
Agentforce	NM Support Agent with the Support Ticket Priority Analysis subagent and the Flow-based Analyze Support Ticket Priority action.
Testing	Five runtime scenarios executed; all passed.
Security	Agent scope restriction and Flow-only actions; no profiles or permission sets are specified.
Version Control	Public GitHub repository Nm-Agentforce-support-Flow.
Maintenance	Keyword, assignment and Agentforce configuration updates; troubleshooting guide.

RESULTS AND DISCUSSION

Results Against Objectives

Table 42: Results Against Objectives

Objective	Result
Classify tickets as High, Medium or Low	Achieved. The Flow classifies the latest ticket from its description; the three levels were confirmed in TC-01 to TC-03.
Retrieve the latest ticket for an Account	Achieved. The Flow retrieves the Account by name and then its latest ticket; the no-ticket case was confirmed in TC-05.
Reduce manual prioritization and assignment effort	Addressed by automation. The Flow determines priority and assigned support level without manual decisions.
Create an urgent handling task for High priority	Configured in the Flow (Subject “Urgent Ticket Handling”, Priority High, Status Not Started).
Use Agentforce to analyze ticket details and trigger the Flow	Achieved. The subagent collects the Account Name, invokes the Flow and presents the outputs (TC-01 to TC-05).

Discussion

The runtime tests show that the solution behaves as configured. Tickets for the three test Accounts were classified High, Medium and Low, each with the expected support level and message, and both exception scenarios were handled by Agentforce as intended.

The behaviour is driven by keyword conditions. This makes it predictable and easy to maintain: the same description always leads to the same result, and the rules can be changed in the Flow. It also bounds what the solution can understand. A ticket that signals urgency in words outside the configured keyword set is classified by the keywords it does contain, or as Low.

The division of work between Agentforce and the Flow keeps the conversational interface separate from the business rules. Agentforce asks for missing input, calls the action and presents the outputs, while the Flow holds the logic. Because the subagent is instructed to act only through the Flow, the same rules apply however the Flow is invoked.

The solution analyzes the latest ticket of an Account, and SLA-related capability is limited to the SLA Breach Risk field and an optional check. These boundaries are described in the limitations and the future enhancements.

ADVANTAGES AND LIMITATIONS

Advantages

Table 43: Advantages

Advantage	Description
Automatic ticket prioritization	The Flow classifies the latest ticket as High, Medium or Low from its description.
Faster identification of critical tickets	High-priority tickets are identified automatically so that agents can focus on urgent cases first.
Reduced manual effort	Ticket analysis, priority setting, task creation and assignment are performed by configured automation.
Conversational access	Support users give an Account Name to Agentforce and receive the Ticket ID, priority, assigned agent and message.
Automatic urgent task creation	High-priority tickets create an “Urgent Ticket Handling” Task.
Support-level assignment	Tickets map to Senior Support Agent, Support Agent Tier 2 or Support Agent Tier 1 according to priority.
Consistent behaviour	The same description is always evaluated against the same configured conditions.
Controlled agent scope	The subagent is limited to ticket analysis and acts only through the Flow.
Declarative and scalable	The solution is built with configuration and no custom code, and Salesforce automation supports increasing ticket volumes.

Limitations

Table 44: Limitations

Limitation	Description
Keyword-based priority logic	Classification depends on the configured keywords: urgent, not working, failure, issue, slow and delay.
Limited contextual understanding	Urgency expressed in other words is not recognized unless the conditions are extended.
Latest ticket only	Only the most recent ticket of an Account is analyzed.
Dependence on accurate data	The Account is found by its Name (the first matching record is used), and a correct Customer lookup and a meaningful Description are needed for a correct analysis.
Flow and Agentforce configuration dependency	Changes to ticket handling require maintenance of the Flow and of the subagent and action configuration.
Limited agent scope	Billing, subscription management and account updates are outside the subagent’s scope.
SLA capability is limited	The SLA Breach Risk field is available and an SLA check is optional; comprehensive SLA monitoring and escalation are not implemented.
Fixed assignment levels	Assignment is to three support levels; additional teams or rules need further configuration. Analytics for priority trends are not part of the implementation.

FUTURE ENHANCEMENTS

Table 45: Future Enhancements

Enhancement	Description
Additional priority indicators	Add more priority indicators and business rules beyond the current keyword set.
Richer Agentforce outputs	Add more variables to the Flow and configure them in the Agentforce action to return more ticket information.
Advanced SLA monitoring and escalation	Extend the SLA Breach Risk field and the optional SLA check into monitoring and escalation.
More support teams and assignment rules	Support additional support teams and more detailed assignment rules.
Analytics	Introduce analytics for priority trends and resolution performance.
Notifications for urgent tickets	Add notifications for urgent tickets, an idea identified during brainstorming.

CONCLUSION & PROJECT OUTCOMES

The Customer Support Ticket Priority Prediction and Automated Assignment System demonstrates how Salesforce Flow and Agentforce can work together to reduce manual support-ticket handling. The solution retrieves the latest ticket for an Account, analyzes its description, classifies priority, creates an urgent task for High-priority cases, returns the assigned support level and presents a clear conversational response. Agentforce provides the conversational interface and the Flow provides the business logic; no independent machine-learning model is involved.

Project Outcomes

- A Support Ticket Intelligence custom object with eleven fields, including an auto-numbered Ticket ID (TKT-{0000}) and lookups to Account, Contact and User.
- An Auto-Launched Flow that retrieves the latest ticket for an Account, classifies priority from the description, assigns a support level, sets a status message and creates an urgent Task for High priority.
- An Agentforce configuration (NM Support Agent, Support Ticket Priority Analysis subagent and Analyze Support Ticket Priority action) that collects the Account Name, invokes the Flow and returns the Ticket ID, priority, assigned agent and message.
- Runtime validation in which all five test scenarios passed: High, Medium and Low priority, a missing Account Name and an Account with no support ticket.
- A public GitHub repository for the project.

Scope Boundaries

- Priority classification is keyword-based and uses the configured conditions.
- SLA support is limited to the SLA Breach Risk field and an optional check; comprehensive SLA monitoring is not implemented.
- Further capabilities, such as analytics and additional assignment rules, are identified as future enhancements.

APPENDIX

Table 46: Appendix

Ref.	Item	Details
A	Custom Object	Support Ticket Intelligence – Support_Ticket_Intelligence__c (Section 3.2, 3.3)
B	Key Salesforce Records	Account, Contact, User, Task (Section 3.4)
C	Flow	Support Ticket Intelligence – Auto-Launched Flow (Phase 4)
D	Agentforce Subagent	Support Ticket Priority Analysis – Support_Ticket_Priority_Analysis (Section 5.3, 5.4)
E	Agentforce Action	Analyze Support Ticket Priority – Flow-based action (Section 5.5)
F	Runtime Test Results	TC-01 to TC-05 (Section 6.2)
G	Project Repository	https://github.com/mohanraj3010/Nm-Agentforce-support-Flow.git (Section 7.2)